

The Role of Hypoxia-Induced Factors in Tumor Progression

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Key Words. Hypoxia · HIF-1 α · Proteome changes · Angiogenesis · Genome changes

LEARNING OBJECTIVES

After completing this course, the reader will be able to:

1. Describe hypoxia-induced mechanisms for cell survival.
2. Discuss hypoxia-induced gene expression.
3. Relate hypoxia and glucose metabolism.

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ABSTRACT

Hypoxia is a common characteristic of locally advanced solid tumors that has been associated with diminished therapeutic response and, more recently, with malignant progression, that is, an increasing probability of recurrence, locoregional spread, and distant metastasis. Emerging evidence indicates that the effect of hypoxia on malignant progression is mediated by a series of hypoxia-induced proteomic and genomic changes activating angiogenesis, anaerobic metabolism, and other processes that enable tumor cells to survive or escape their oxygen-

deficient environment. The transcription factor hypoxia-inducible factor 1 (HIF-1) is a major regulator of tumor cell adaptation to hypoxic stress. Tumor cells with proteomic and genomic changes favoring survival under hypoxic conditions will proliferate, thereby further aggravating the hypoxia. The selection and expansion of new (and more aggressive) clones, which eventually become the dominant tumor cell type, lead to the establishment of a vicious circle of hypoxia and malignant progression. *The Oncologist* 2004(suppl 5):10-17

INTRODUCTION

Solid tumors comprise approximately 90% of all known cancers [1]. They develop from a single mutated cell and lead to significant morbidity and mortality, either by invading normal tissue or by metastasizing to vital organs, such as the liver, lung, or brain. The process of tumor progression (i.e., proliferation, local invasion, and distant metastasis) is characterized by rapid cellular growth accompanied by alterations of the microenvironment of the tumor cells. To a large extent, the alterations in the cellular

microenvironment are due to an inadequate oxygen (O₂) supply and the resultant hypoxia or even anoxia [2, 3].

To grow beyond a diameter of approximately 1 mm, newly developing tumors must form their own vascular network and blood supply, which they accomplish either by incorporating preexisting host vessels or by forming new microvessels through the influence of tumor angiogenesis factors [2, 4]. However, the newly formed vascular network differs greatly from that found in normal tissue, typically displaying a broad range of structural and functional

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Received August 19, 2004; accepted for publication September 5, 2004. ©AlphaMed Press 1083-7159/2004/\$12.00/0

abnormalities, including dilations, incomplete or absent endothelial linings and basement membranes, leakiness, irregular and tortuous architecture, arteriovenous shunts, blind ends, and a lack of contractile wall components and pharmacological/physiological receptors [2]. These abnormalities lead to irregular and sluggish blood flow, thereby diminishing the delivery of O₂ (and nutrients) to the tumor cells, with the resultant development of hypoxic or even anoxic areas. The oxygenation status of the tumor can be worsened further by increases in diffusion distances, which occur when the tumor cells spread beyond the distance that allows adequate delivery of O₂ by the blood vessels (>70 μ m) [1, 2, 5]. Additionally, diminished tumor oxygenation and subsequent hypoxia can be induced or exacerbated by a reduced O₂ transport capacity of the blood due to the presence of cancer-related or cancer treatment-induced anemia.

For many years, tumor hypoxia has been recognized as a potential therapeutic problem because of its adverse impact on the effectiveness of radiation therapy. However, hypoxia has recently emerged as a major factor that influences tumor proliferation and malignant progression [6]. Although some of the effects of hypoxia negatively impact tumor cell growth [7], they may, antithetically, lead to hypoxia-driven responses that enhance malignant progression and aggressiveness, ultimately resulting in increased resistance to therapy and a poor long-term prognosis. Malignant progression associated with tumor hypoxia appears to be mediated by several mechanisms, including changes in gene expression, inactivation of suppressor genes or activation of oncogenes, genomic instability, and clonal selection.

HYPOXIA-INDUCED MECHANISMS FOR CELL SURVIVAL, INVASION, AND METASTASIS

Hypoxia (oxygen tension [pO₂] <7 mmHg) can induce changes in the proteome of tumor cells that lead to impaired growth or to cell death, including cell-cycle arrest, differentiation, apoptosis, and necrosis [8-12]. Alternatively, however, hypoxia can induce proteomic changes that allow the tumor cells to successfully adapt to or overcome their O₂- and nutrient-deprived state and to survive in or escape from their hostile environment. This is accomplished through hypoxia-stimulated angiogenesis, glycolysis, inhibition of apoptosis, and upregulation of growth factors (e.g., platelet-derived growth factor-B [PDGF-B], transforming growth factor beta [TGF- β], insulin-like growth factor-2 [IGF-2], epidermal growth factor [EGF]) and other proteins involved in tumor invasiveness (e.g., urokinase-type plasminogen activator). Systemic responses leading to an elevation in the hemoglobin level, and thus improvement in the O₂ transport capacity of the blood, can support the local mechanisms mentioned within tumors (e.g., through

activation of the genes for erythropoietin, transferrin, and transferrin receptors) [6, 13]. Additionally, hypoxia may induce downregulation of adhesion molecules, thereby facilitating tumor cell detachment [14, 15]. Many of these hypoxia-inducible genes are controlled by hypoxia-inducible factor 1 (HIF-1) (see below).

CHANGES IN GENE EXPRESSION: HIF-1 AND OTHER FACTORS

Cells that are poorly oxygenated (pO₂ <7 mmHg) display a series of adaptive responses that allow for survival and continued proliferation. Among these, changes in the expression of genes for erythropoietin, the angiogenic vascular endothelial growth factor (VEGF), transferrin receptors, and other proteins allow for the development of a more effective O₂ (and nutrient) supply. Another group of genes involved in this adaptive response controls metabolic pathways that can meet the cellular energy requirements (e.g., glycolytic enzymes and glucose transporters). Expression of the genes for most of these proteins is regulated by HIF-1 α . This transcription factor was first identified by *Semenza* and colleagues as a regulator of hypoxia-induced erythropoietin expression [16-18] and has since been demonstrated to regulate the expression of more than 30 target genes (Table 1). These genes also play roles in tumor progression (i.e., proliferation, invasion, and metastasis), thereby contributing to tumor aggressiveness (Fig. 1) [19]. Other factors involved in the regulation of O₂-dependent transcription are nuclear factor kappa B (NF- κ B) and activator protein-1 (AP-1) (see below).

Table 1. Known HIF-1 target genes (gene products) [19]

Adenylate kinase 3	IGF-2
α_{1B} -adrenergic receptor	IGF binding protein 1
Adrenomedullin	IGF binding protein 3
Aldolase A	Lactate dehydrogenase A
Aldolase C	Nitric oxide synthetase 2 (NOS 2)
Carbonic anhydrase IX	p21
Carbonic anhydrase XII	p35srj
Coeruloplasmin	Phosphofructokinase L
Endothelin-1 (ET-1)	Phosphoglycerate kinase 1
Enolase 1 (ENO1)	Plasminogen activator inhibitor-1
Erythropoietin (EPO)	PDGF-B
GLUT-1	Pyruvate kinase M
Glyceraldehyde phosphate dehydrogenase	Transferrin receptor
Heme oxygenase 1	TGF- α
Hexokinase 1	VEGF
Hexokinase 2	Flt-1

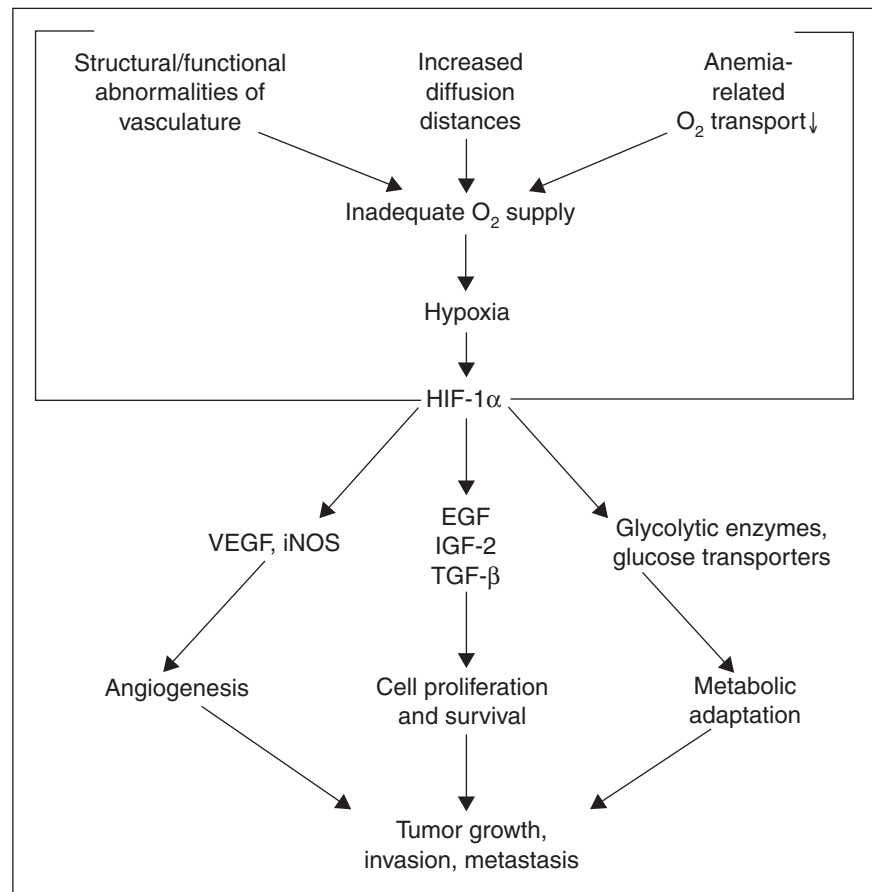
Figure 1. Expression of HIF-1 in human cancer: causes and consequences.

HIF-1

HIF-1 is a heterodimer comprising HIF-1 α and HIF-1 β subunits, both of which are basic helix-loop-helix transcription factors [17, 20]. HIF-1 β (ARNT) is a nuclear protein that is constitutively expressed and is independent of O₂ tension [21]. HIF-1 α , in contrast to HIF-1 β , is a cytoplasmic protein responsive to O₂ levels. In well-oxygenated cells, HIF-1 α is continuously degraded by the ubiquitin-proteasome system. This degradation process takes place only when certain conserved prolyl residues of HIF-1 α are hydroxylated, a modification requiring O₂-dependent enzyme activity [22]. Only HIF-1 α containing modified prolyl sites binds to the von Hippel-Lindau protein, which is the recognition component of an E3 ubiquitin ligase that finally targets HIF-1 α for proteasomal degradation. Under hypoxic conditions, HIF-1 α subunits translocate to the nucleus, where they heterodimerize with HIF-1 β subunits. The resultant product is an active HIF-1 protein that binds to specific hypoxic response elements present in target genes, ultimately activating transcription of these genes (Fig. 2), which encode for erythropoietin, VEGF, various glycolytic enzymes, transferrin, and a variety of other proteins essential for systemic, local, and intracellular homeostasis. Importantly, the vast majority of these gene products are overexpressed in human tumor cells [19], suggesting that the HIF-dependent transcriptome changes are important in tumor pathophysiology. Overall, these adaptive responses to low O₂ levels serve as a compensatory mechanism for increasing delivery of O₂ (and nutrients) for any body cells with an inadequate O₂ supply. However, for hypoxic tumor cells, these adaptive responses can additionally favor cell survival, further expansion, and metastasis, as outlined below [19-26].

HIF-1, VEGF, AND ANGIOGENESIS

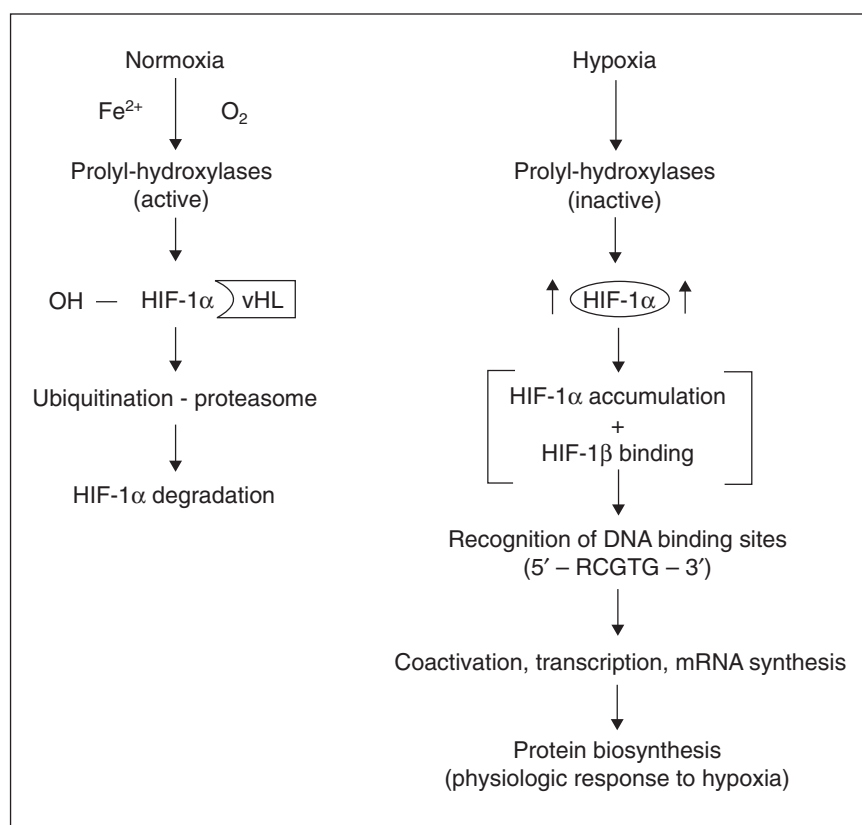
Angiogenesis is the process by which new blood vessels develop from existing vasculature, thereby providing a



principle mechanism for the maintenance of an adequate blood flow in expanding cell populations, including those of tumor tissue. In a rapidly growing tumor, O₂ demand increases and O₂ delivery decreases, primarily because of: A) insufficient blood supply (at least to some tumor areas) and B) increasing diffusion distances between the blood vessels and the O₂-consuming cells [2, 20]. This leads to hypoxia in the expanding tumor mass, triggering events that stimulate angiogenesis in an effort to ameliorate the hypoxic condition. In tumor tissue, the ability to induce angiogenesis is associated with the development of an aggressive phenotype, as metastatic cells have more opportunity to enter the circulation in a well-vascularized tumor and thereby escape their hostile environment [25, 27].

One of the most potent stimulators of angiogenesis is VEGF, which is essential for the proliferation and migration of vascular endothelial cells, thereby enabling the formation of new blood vessels [10, 24, 28, 29]. Production of VEGF is driven by hypoxia via transcription activation of the VEGF gene by HIF-1 [30]. The basic importance of HIF-1 in the angiogenic process has been demonstrated in several experimental and clinical studies [31]. Carmeliet *et al.* reported a reduced hypoxic induction of VEGF in vitro in mouse embryonic stem cells with inactivated HIF-1 α genes [32].

Figure 2. Regulation of HIF-1 α by cellular O₂ level. O₂ determines the subjection of HIF-1 α to protein hydroxylation. Under normoxic conditions, ubiquitination of HIF-1 α targets the subunit for proteasome degradation. Under hypoxic conditions, HIF-1 β dimerizes with HIF-1 α , and the active HIF-1 dimer binds to hypoxia response elements containing the core recognition sequence 5'-RCGTC-3' and then recruits coactivator molecules, resulting in the formation of an increased transcription initiation complex and mRNA synthesis, leading ultimately to the biosynthesis of proteins that mediate responses to hypoxia [23].



Ryan *et al.*, in *in vivo* studies, found that HIF-1 α ^{-/-} embryonic stem-cell-derived tumors had fewer blood vessels and impaired hemodynamics within the tumor mass, and further, that HIF-1 α knockout mice died in utero with a complete lack of cephalic vasculature as a consequence of disrupted angiogenesis [3, 33]. Additionally, VEGF has been shown to stimulate migration of macrophages by activation of the VEGF receptor (Flt-1). Macrophages produce several angiogenic factors, including VEGF and tumor necrosis factor alpha (TNF- α) [34, 35]. At the clinical level, the results of the majority of over a dozen studies comprising more than 3,500 patients generally speak in favor of an independent prognostic impact of VEGF expression regarding relapse-free and overall survival. Additionally, VEGF expression may be predictive of the anatomical site of first recurrence [25].

In addition to VEGF, other angiogenesis-related gene products and receptors are regulated by HIF-1, including PDGF-B, VEGFR-1, endothelin-1, inducible nitric oxide synthetase (iNOS), monocyte chemotactic protein, adrenomedullin, and EGF. Several of these, including iNOS, endothelin-1, heme oxygenase 1, and adrenomedullin, have been shown to play roles in the regulation of local blood flow by the modulation of vascular tone [36]. Thus, it appears that HIF-1 not only mediates angiogenesis by VEGF induction but also influences tumor blood flow by more complex mechanisms involving target genes playing a role in vessel tone.

Yet another mechanism for stimulation of tumor angiogenesis is induction of HIF-1 and VEGF subsequent to somatic mutation. One example of this is seen in the loss of

p53 tumor-suppressor activity either by direct mutational inactivation or by overexpression of mouse double minute, a ubiquitin protein ligase involved in the degradation of p53. Loss of p53 activity results in decreased hypoxia-mediated apoptosis, possibly increased HIF-1 α expression [37], and a subsequent increase in HIF-1-mediated transactivation of VEGF and other target genes, thereby facilitating tumor angiogenesis.

HIF-1 AND GLUCOSE METABOLISM

In contrast to normal cells, tumor cells characteristically display a relatively high glycolytic rate, even when growing in the presence of O₂. Under normoxic conditions, cells generate ATP via oxidative phosphorylation. However, in the expanding tumor mass, which is generally characterized by a limited O₂ supply and a high glucose consumption rate, anaerobic glycolysis can become the predominant pathway of ATP generation [2, 36]. This metabolic shift appears to be regulated by HIF-1 (Fig. 3). Enzymes—including aldolase A, phosphoglycerate kinase 1, and pyruvate kinase M—are induced by HIF-1 *in vitro*, and lactate dehydrogenase is induced by HIF-1 α in breast carcinoma lines [38]. The efficacy of the glycolytic response is enhanced by overexpression of other proteins, including glucose transporters (e.g., glucose transporter 1 [GLUT-1]), which facilitate glucose uptake by the cells, and by hexokinase, which enhances the capacity of

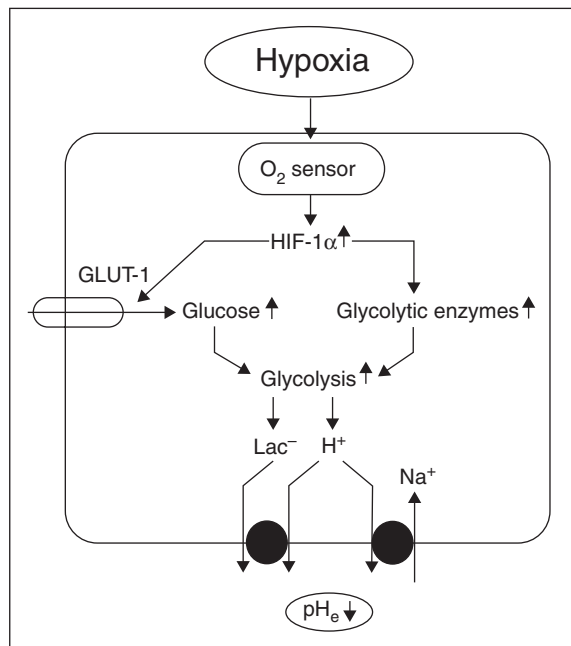


Figure 3. HIF-1-mediated switch from aerobic to anaerobic metabolism in hypoxic tumors for energy preservation. The activation of genes for glucose transporters (GLUT-1) and glycolytic enzymes results in an increased glycolytic rate. H^+ ions produced are preferentially exported via a lactate/ H^+ symporter and a Na^+/H^+ antiporter, leading to a decrease in extracellular pH [44].

tumor cells to catabolize glucose at higher metabolic rates, thereby increasing the production of precursors needed for cell growth and maintaining high ATP production under conditions of O_2 deficiency [39]. Thus, HIF-1-induced adaptive responses not only provide for VEGF-mediated angiogenesis, but also ensure that the energy requirements of the cells are met, thereby allowing their survival in a hostile environment.

HIF-1 INDEPENDENT PATHWAYS

Although HIF-1 seems to play a pivotal role in hypoxic response, other hypoxia-regulated transcription factors do exist. For example, NF- κ B can also be activated by hypoxia [40]. Activation of NF- κ B leads to transcription of target genes such as those encoding proinflammatory cytokines (e.g., interleukins 6 and 8, TNF- α) and cyclooxygenase-2 (COX-2). COX-2 has angiogenic and growth-stimulatory properties, and is able to activate the genes for urokinase-like plasminogen activator and matrix metalloproteinase-2, both of which are associated with tumor invasiveness. NF- κ B has also been shown to play an important role in apoptosis regulation since it leads to overexpression of the antiapoptotic factor bcl-2.

AP-1 has also been identified as a hypoxia-inducible transcription factor. Prolonged AP-1 activation by hypoxia

may depend on HIF-1 α , with both of these cooperating in the transactivation of target genes [41]. However, such interactions are still not completely understood and require further studies aimed at elucidating more details.

GENOMIC INSTABILITY

The tumor microenvironment is considered hostile, being characterized by areas of chronic or transient hypoxia, low pH, nutrient deprivation, and energy depletion. In a classic study, Reynolds and colleagues examined the consequences of tumor growth under these conditions, using a tumorigenic cell line carrying a recoverable, chromosomally based lambda phage shuttle vector designed to identify mutations without the need for a genetic selection of mutant cells [42]. The cells were grown concurrently either in culture or as tumors in nude mice. The frequency of mutations in the cells within the murine tumors was found to be five times that of the comparator cultured cells (9.3×10^{-5} versus 1.8×10^{-5} , respectively; $p < 0.0001$). Moreover, the mutation patterns of the two cell groups differed, with the tumor-grown cells displaying significantly more deletions and transversions than those grown in culture. Particularly noteworthy is the finding that exposure of cultured cells to hypoxic conditions produced an elevated mutation frequency and a mutation pattern similar to those observed in the tumor-grown cells. These findings suggest that the type of genetic instability found in malignant tumors may in part be the consequence of specific mutagenic properties of the hypoxic microenvironment [43].

HYPOXIA-INDUCED GENOME CHANGES AND CLONAL SELECTION

Accumulating evidence suggests that hypoxia may lead to malignant progression by means of genomic changes in the tumor cells and clonal selection. Both of these actions have been associated with tumor cell pO_2 values ≤ 0.7 mm Hg [44] (Fig. 4). Hypoxia, with or without reoxygenation, promotes genomic instability through point mutations, gene amplification, and chromosomal rearrangement [45].

Point mutations may develop in tumor cells exposed to hypoxia and reoxygenation through several mechanisms, including insufficient DNA repair, errors in DNA replication, or both [42, 46]. Metabolic damage to DNA bases may also play a role in point mutations, since a hypoxia-reoxygenation sequence may cause oxidative damage. Such damage has the potential to lead to various pyrimidine- and purine-derived lesions in DNA. The most abundant of these is the generation of 8-hydroxyguanine, which has been shown to mispair with adenine and lead to G:C to T:A transversions [47, 48].

Several studies have demonstrated that hypoxia followed by reoxygenation can lead to gene amplification [49], which,

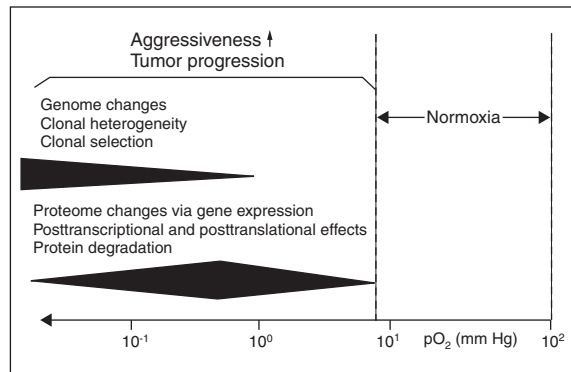


Figure 4. Approximate “critical” O_2 levels in solid tumors leading to alterations in gene expression and posttranslational and post-transcriptional modulations resulting in proteomic changes, and approximate critical O_2 levels furthering persistent genomic changes and clonal selection. These hypoxia-mediated changes can promote tumor aggressiveness and malignant progression. Reprinted from Vaupel et al. [5].

together with chromosomal rearrangements, can be caused by DNA strand breaks or decreased repair of DNA strand breaks [45]. The strand breaks may occur as a result of increased expression of endogenous endonuclease [50]. Hypoxia-induced point mutations, chromosomal rearrangements, and gene amplification may, in turn, promote development of metastatic disease by several mechanisms, including inactivation of metastasis suppressor genes or increased expression of oncogenes involved in the metastatic process, for example, genes encoding for angiogenesis and growth factors.

The overall effect of hypoxia-induced mutation and gene amplification is an increase in the number of gene variants. It has also been suggested that hypoxia exerts a strong selection pressure on malignant cells [5, 6, 51, 52]. Thus, any malignant cells with proteomic or genomic adaptive changes favoring survival under hypoxic conditions (e.g., decreased capacity for cell-cycle arrest, differentiation, or apoptosis, or increased angiogenic potential) will have selection advantages over nonadapted cells. The progeny of the adapted cells will increase at a greater rate than those of the nonadapted cells and eventually will become the dominant cell subpopulation within the tumor. Moreover, these cells are likely to have more favorable traits related to invasion, metastasis capability, and aggressiveness, providing the basis for the clinical findings of increased locoregional spread, distant tumor metastasis, and treatment resistance in advanced disease. Additionally, hypoxia-mediated clonal selection of tumor cells with genomic changes leading to apoptotic insensitivity, and possibly increased angiogenic potential, further aggravates tumor hypoxia and establishes a vicious circle of hypoxia and malignant progression that is considered a

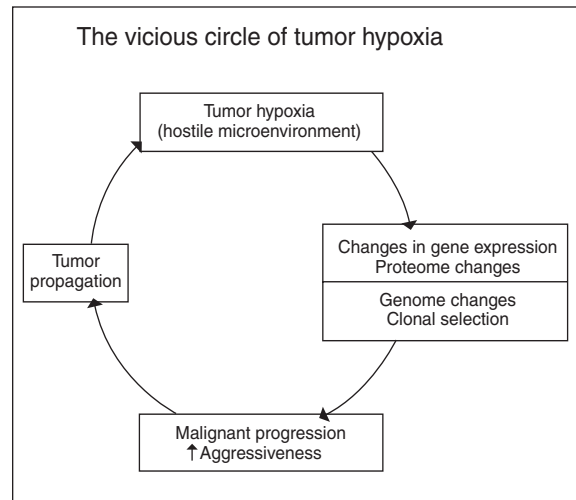


Figure 5. Schematic displaying the vicious circle of tumor hypoxia and malignant progression [6].

pivotal biological mechanism of advanced (and often incurable) disease (Fig. 5) [52].

REOXYGENATION AND MALIGNANT PROGRESSION

Results of several preclinical studies have provided evidence that hypoxia, with or without reoxygenation, may result in malignant progression and poor prognosis. In the Reynolds *et al.* study discussed above, the frequency and pattern of mutations in hypoxically cultured cells were similar to those observed in the tumor-grown cells [42]. The mutation frequency of the cultured cells continued to rise with repeated exposure to hypoxia followed by reoxygenation, suggesting impairment of cellular repair capabilities. It has been suggested that repeated hypoxia-reoxygenation cycles may function as a mutagenic force by increasing the levels of superoxides and other O_2 radicals [53]. Cycles may also lead to chromosomal rearrangements and gene amplification [43]. As stated in a lecture given by P.W. Vaupel, M.D. (1994), at the Ernst Schering Research Foundation in Berlin, it is well recognized in the clinical setting that patients receiving blood transfusions experience intermittent hypoxia and reoxygenation [54, 55]. Reoxygenation-related increases in free radical formation can, in turn, activate stress response genes, such as heat shock protein 70 (which is an effective inhibitor of apoptosis), or stress-response transcription factors, such as NF- κ B (which regulates numerous genes including VEGF), potentially leading to malignant progression.

SUMMARY AND CONCLUSIONS

Because of its demonstrated impact on malignant progression and therapeutic response, leading to a poor

long-term disease outcome, tumor hypoxia is a growing concern in the oncology setting. Results of preclinical and clinical investigations during the past decade have established that tumor hypoxia may promote malignant progression by several mechanisms, including an increased expression of transcription factors and gene products involved in tumor propagation and induction of genomic instability (e.g., point mutations, deletions, and gene amplification). In those investigations, the transcriptional factor

HIF-1 has emerged as a major regulator of adaptive processes (including angiogenesis) that can support tumor cell survival, proliferation, invasion, and metastatic spread. Also, it has been shown that hypoxia can enhance malignant progression and increase aggressiveness through clonal selection. Therefore, in developing treatment strategies for cancer patients, it is reasonable to consider approaches aimed at ameliorating tumor hypoxia in an effort to maximize the effects of cancer therapy.

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